EXP.No 5 VERIFICATION OF NORTON'S THEOREM
DATE 24/2/26

AIM:
To verify Norton's theoram practically and theoretically for the given DC circuit

APPARATUS REQUIRED:

SL.NO	APPARATUS	SPECIFICATION	QUANTITY
1	Regulated Power supply	(0-30V)	1
2	Voltmeter	(0-30V) MC	1
3	Ammeter	(0-10mA) MC	1
4	Resistors	470 Ω, 560 Ω, 1K Ω	2, 1, 1
5	Bread Board	–	1
6	Multimeter	–	1

PROCEDURE:
1) Make the connection as per the Circuit Diagram 1
2) Vary the RPS and set an input voltage of 10V.
3) Note down the voltmeter reading (Vi) and ammeter reading (I_L) in TC.1
4) Switch off the supply and make connections for Circuit Diagram 2.
5) Measure the Thevenin's resistance R_{TH} = Norton's resistance R_N
6) Switch off the supply and make connections for Circuit Diagram: 3.
7) Set an input voltage of 10V in the RPS and note down the voltmeter readings Vi and V_{TH} (= V_{OC}) in Tabular column: 3
8) Switch off the supply and make connections for circuit diagram 4
9) Set an input voltage of 10V in the RPS and note down the voltmeter reading Vi and ammeter reading I_N (= I_{SC}) in Tabular Column 4.
10) Draw the thevenin's equivalent circuit and Norton's equivalent circuit

Tabular column : 1

To measure I_L

Vi (volts)	I_L (amps)
10V	2.67mA

Tabular Column : 2.

To measure R_{TH} or R_N

Vi (volts)	Rin (Ω)
10V	7.16 Ω

Tabular Column : 3

To measure I_N or I_{SC}.

Vi (volts)	In (amps)
10 V	5.94mA.

Model Calculation :-

Verification of Norton's theorem :- $I_L = I_N * R_N / (R_N + R_L)$

Theoretical calculation of I_L, I_N and Given circuit.

$$R_T = 470 + \frac{470 \times 560}{470 + 560} \Rightarrow 470 + 255.5$$

$$R_T = 725.53\Omega$$

$$I_{Ti} = \frac{10}{725.52} = 0.013A$$

$$I_N = 0.013 \times \frac{470}{470 + 560} \Rightarrow 5.9mA$$

$$R_N = 560 + \frac{470 \times 470}{470 + 470} = 795\Omega$$

$$I_L = \frac{I_N \times R_N}{(R_N + R_L)}$$

$$= \frac{5.9 \times 10^{-3} \times 795}{795 \times 1000} = 2.01MA$$

o/p verified

$$V_1 = 2.61 \times 10^{-3} \times 10^3$$

$$V_L = 2.61V$$

Verification

$$I_L = \frac{I_N * R_N}{R_N + R_L} = 2.45mA$$

11) Calculate the I_L value using the formula thevenin's theoren.

$$I_L = V_{Th} / (R_{Th} + R_L)$$

Norton's Theorem

$$I_L = I_n * R_N / (R_N + R_L)$$

12) Theoretically verify the Norton's theorem.

Mark split-up

SNo	Description	Marks allotted	Marks obtained.
1	Experimental setup/ connection	20	20
2	Observation / Reading	20	20
3	Calculation / Results	20	20
4	Viva	20	20
5	Record	20	20
		Total	100

100

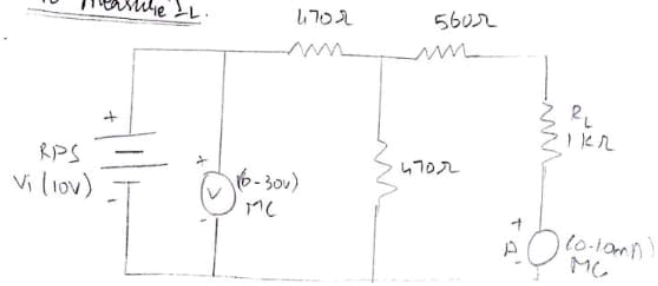
7/3/26

RESULT:

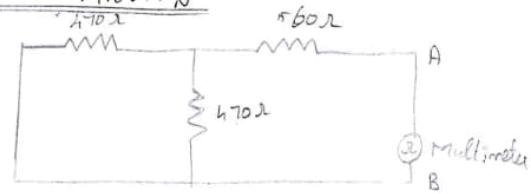
Thus Norton's theorem is verified practially and theoretically.

CIRCUIT DIAGRAM:

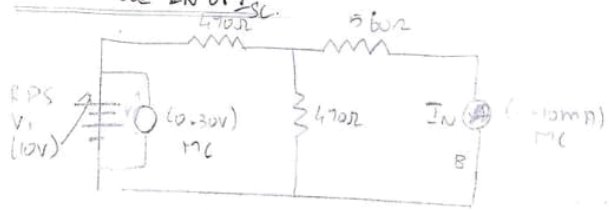
To measure I_L.



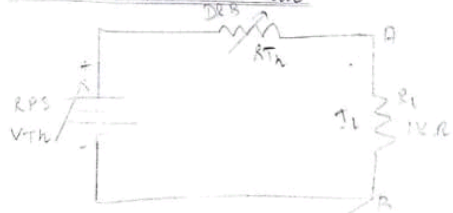
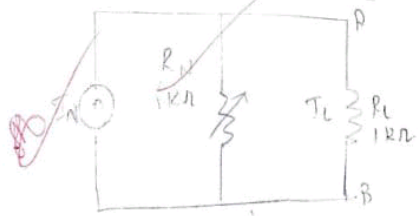
To measure R_{Th} or R_N



To measure I_N or I_{SC}.



Thevenin's equivalent circuit



Norton's equivalent